

CHE 435/525 *Process Systems Analysis and Control*

Lecture 8: **Second-Order Systems and their Characteristics**

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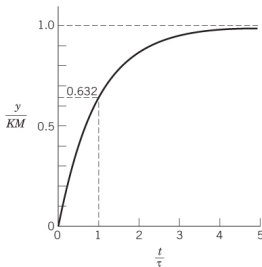
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Recapture: First-order systems and integrating systems

- Step response of $G(s) = K/(\tau s + 1)$

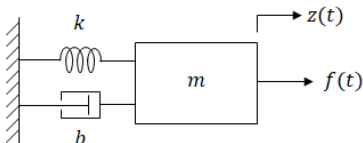


- ▶ In practice: From the step response curve, determine the t.f. (K and τ)
- First-order system: **“Capacitance and resistance”**
- Integrating system: Output is the **accumulation** of input

Learning objectives of this lecture

- ① Explain what kind of physical processes are **second-order systems**
- ② Discriminate the behaviors of **underdamped, critically damped, and overdamped** second-order systems
- ③ Indicate the **characteristic quantities** on the response curves and relate them to the model parameters

Example 1: Mass-spring-damper system



- Model

$$m \frac{d^2 z}{dt^2} = -b \frac{dz}{dt} - kz + f, \quad \text{or} \quad \underline{\frac{dz}{dt} = v, \quad m \frac{dv}{dt} = -bv - kz + f}$$

- ▶ mass $\times$ acceleration = friction + elastic force + external forcing

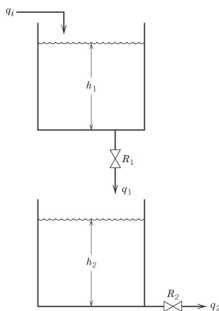
- Transfer function from $u = f$ (forcing) to $y = z$ (displacement)?

$$G(s) = \frac{1}{ms^2 + bs + k} \quad \left(= \frac{1/k}{(m/k)s^2 + (b/k)s + 1} \right)$$

- What causes the system to be “second-order”?

- ▶ It's a mechanical system with inertia (mass)!

Example 2: Two tanks in series



- Model should look like

$$\frac{Q_1(s)}{Q_i(s)} = \frac{1}{\tau_1 s + 1}, \quad \frac{Q_2(s)}{Q_1(s)} = \frac{1}{\tau_2 s + 1}$$
$$\Rightarrow \frac{Q_2(s)}{Q_i(s)} = \frac{1}{\tau_1 \tau_2 s^2 + (\tau_1 + \tau_2)s + 1}$$

- Discuss: What causes the system to be “second-order”?
 - Indirect effect of input on output, through an intermediate variable

Example 3: Non-isothermal CSTR [from Lecture 6]

CSTR [Seborg Ex.4.10]

A CSTR with first-order reaction $A \rightarrow B$ using cooling water:

$$V \frac{dc_A}{dt} = q(c_{Ai} - c_A) - k_0 e^{-E/RT} c_A V$$

$$\rho V C_p \frac{dT}{dt} = \rho q C_p (T_i - T) + k_0 e^{-E/RT} c_A V (-\Delta H_R) + UA(T_c - T)$$

Assume $c_{Ai} = \text{const.}$ Find the transfer function $C'_A(s)/T'_c(s)$.

- Linearize: $\frac{dc'_A}{dt} = a_{11}c'_A + a_{12}T'$, $\frac{dT'}{dt} = a_{21}c'_A + a_{22}T' + bT'_c$.
- L.T. and eliminate $T'(s)$:

$$\frac{C'_A(s)}{T'_c(s)} = \frac{a_{12}b}{s^2 - (a_{11} + a_{22})s + (a_{11}a_{22} - a_{12}a_{21})}.$$

- Discuss: What causes the system to be “second-order”?
 - Coupling of two variables in the process: c_A and T

Second-order system

- Physical picture: A process that must be described by **two (first-order) ODEs**, which allows an **indirect** effect of input on output
- Form of t.f.

$$G(s) = \frac{K}{\tau^2 s^2 + 2\tau\zeta s + 1}.$$

- ▶ $K = G(0)$ is the **static gain**, τ is the **time constant**
 - ▶ ζ is called the **damping coefficient**, **dimensionless**
 - ★ When $\zeta > 1$: **overdamped**
 - ★ When $\zeta = 1$: **critically damped**
 - ★ When $\zeta < 1$: **underdamped**
- Exercise: For two-tank-in-series, $G(s) = \frac{1}{\tau_1\tau_2s^2 + (\tau_1 + \tau_2)s + 1}$, find τ and ζ . Is it underdamped, overdamped, or critically damped?

Calculating step response

- Step input $U(s) = M/s$

$$Y(s) = \frac{KM}{s(\tau^2 s^2 + 2\tau\zeta s + 1)}$$

- The qualitative behavior of $y(t)$ depends on the roots of $(\tau^2 s^2 + 2\tau\zeta s + 1)$.
 - ▶ Depending on the discriminant $\Delta = 4\tau^2\zeta^2 - 4\tau^2 = 4\tau^2(\zeta^2 - 1)$, two roots can be **two real numbers** or **a pair of complex conjugates**.
 - ▶ Two roots satisfy $s_1 + s_2 = -2\zeta/\tau$ and $s_1 s_2 = 1/\tau^2$ (How do I know?)
 - ▶ So, the two roots are **two different negative real numbers** (if $\Delta < 0$), **two identical negative real numbers** (if $\Delta = 0$), or **a pair of complex conjugates with a negative real part** (if $\Delta > 0$).

Overdamped & critically damped 2nd-order systems

- Case 1: $\zeta > 1$ (**overdamped**), two distinct negative real roots

$$y(t) = KM \left[1 - \frac{\tau_1 e^{-t/\tau_1} - \tau_2 e^{-t/\tau_2}}{\tau_1 - \tau_2} \right], \text{ where } \tau_1, \tau_2 = \frac{\tau}{\zeta \pm \sqrt{\zeta^2 - 1}}$$

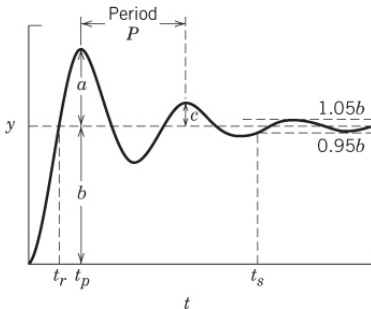
- Case 2: $\zeta = 1$ (**critically damped**), two identical negative real roots

$$y(t) = KM \left[1 - \left(1 + \frac{t}{\tau} \right) e^{-t/\tau} \right]$$

Underdamped 2nd-order systems

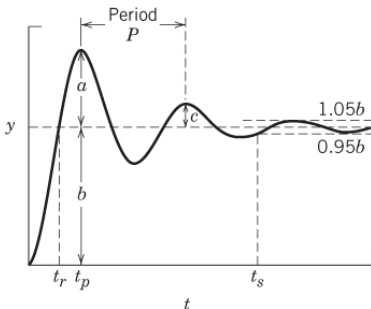
- Case 3: $\zeta < 1$ (**underdamped**), a pair of complex conjugate roots

$$\frac{y(t)}{KM} = 1 - \frac{e^{-\zeta t/\tau}}{\sqrt{1-\zeta^2}} \sin\left(\frac{\sqrt{1-\zeta^2}}{\tau}t + \varphi\right), \text{ where } \varphi = \arcsin \sqrt{1-\zeta^2}$$



- Oscillatory response!!

Underdamped 2nd-order systems



- Characteristics

- ▶ Time to first peak $t_p = \pi\tau / \sqrt{1 - \zeta^2}$
- ▶ Period of oscillations $P = 2\pi\tau / \sqrt{1 - \zeta^2} = t_p$
- ▶ Overshoot $a/b = \exp\left(-\pi\zeta / \sqrt{1 - \zeta^2}\right)$
- ▶ Decay ratio $c/a = \exp\left(-2\pi\zeta / \sqrt{1 - \zeta^2}\right) = \text{Overshoot}^2$
- ▶ Question [Seborg Ex. 5.16]: Settling time $t_s = ?$

Exercise: Analysis of a manometer tube

Manometer model [Seborg Ex.5.10]

Response of liquid level in a manometer leg h to change in pressure p :

$$\frac{d^2 h'}{dt^2} + \frac{6\mu}{R^2 \rho} \frac{dh'}{dt} + \frac{3g}{2L} h' = \frac{3}{4\rho L} p'(t)$$

where R : radius of tube, ρ : density, μ : viscosity, L : length of the liquid column. [** The model can be found at: <https://jckantor.github.io/CBE30338/03.08-Manometer-Models-and-Dynamics.html>]

- ① Find K, τ, ζ in the standard form.
- ② For what values of physical constants does the manometer NOT oscillate?
 - ▶ Answer: $\frac{8\mu^2 L}{3R^4 \rho^2 g} \geq 1$, i.e., $\frac{gR^3}{(\mu/\rho)^2} \cdot \frac{R}{L} \leq \frac{8}{3}$. [**The first dimensionless number is known as “Galileo number” in fluid mechanics.]